Capstone Video Scripts & Abstracts

LEVIS KINYUA MURITHI • IGS 4101 / IGS 4202

1. Innovation in IT: The AgriTrack Application

TRI-LANGUAGE ABSTRACTS

English: This presentation showcases AgriTrack, an innovative digital marketplace and farm management tracker. It eliminates predatory middlemen by connecting smallholder farmers directly to commercial buyers, utilizing React Native and Kotlin Multiplatform for robust, offline-capable supply chain tracking.

Kiswahili: Wasilisho hili linaonyesha AgriTrack, soko la kidijitali na mfumo wa usimamizi wa mashamba. Inaondoa madalali wanyonyaji kwa kuunganisha wakulima wadogo moja kwa moja na wanunuzi wa kibiashara, ikitumia React Native na Kotlin Multiplatform kwa ufuatiliaji thabiti wa ugavi unaofanya kazi hata bila mtandao.

Kikuyu: Uhoro uyu uratariria AgriTrack, soko ya thimu na mutaratara wa kuramata migunda. Niweheragia mabatirania mairigani mbere ya kugwataniria arimi anyinyi na aguri a biashara, ihuthirite mitaratara ya React Native na Kotlin Multiplatform nigutiga ukinyiria wa indo uri mwega o na kuri hatiri mutandao.

20-MINUTE VIDEO SCRIPT OUTLINE

- `[0:00 - 3:00]` **Introduction:** Hook the viewer, introduce AgriTrack, and explain the relevance of agricultural technology in the modern African context.
- `[3:00 - 10:00]` **Deep Dive/Core Analysis:** Explain the current challenges of predatory middlemen and post-harvest losses using real-world farming data.
- `[10:00 - 16:00]` **The IT Solution:** Walk through the AgriTrack architecture. Discuss how offline-first sync (WatermelonDB) and cross-platform access bridge the connectivity gap.
- `[16:00 - 20:00]` **Conclusion & Call to Action:** Summarize the economic empowerment AgriTrack provides and emphasize the need for scalable tech solutions in agriculture.

2. African Culture & IT: Mwethya and Open Source

TRI-LANGUAGE ABSTRACTS

English: This presentation explores the African tradition of 'Mwethya' (communal labor) and parallels it with modern open-source software development in Information Technology. It highlights how collective effort drives innovation and emphasizes the critical importance of gender awareness, ensuring women hold equal leadership roles in both agriculture and the tech industry.

Kiswahili: Wasilisho hili linachunguza utamaduni wa Kiafrika wa 'Mwethya' (kazi ya ushirika) na kuulinganisha na uundaji wa programu huria katika Teknolojia ya Habari. Inaangazia jinsi juhudi za pamoja zinavyochochea ubunifu na kusisitiza umuhimu wa usawa wa kijinsia, kuhakikisha wanawake wanashikilia nafasi sawa za uongozi katika kilimo na tasnia ya teknolojia.

Kikuyu: Uhoro uyu urathuthuria mutugo wa Kiafrika wa 'Mwethya' (wiruti wa hamwe) na kuuringithania na mutaratara wa gwaka programu cia thimu cia andu othe (open-source). Nionanagia uria unyitaniri urehaga ugunduzi na kuheana hinya hari kumenya maundu ma muturire wa atumia na athuri, tigirira atumia mari na iturwa cia utongoria kurima na githomo kia tekinoloji.

20-MINUTE VIDEO SCRIPT OUTLINE (CULTURE & IT)

- `[0:00 - 3:00]` **Introduction:** Introduce the concept of Ubuntu and the specific practice of 'Mwethya' (Ndungata) in traditional communal farming.
- `[3:00 - 10:00]` **Deep Dive/Core Analysis:** Analyze how Mwethya functions socially and economically. Introduce a discussion on historical gender roles in agriculture versus the need for modern gender equity.
- `[10:00 - 16:00]` **The IT Connection:** Draw direct parallels between communal farming and open-source software development. Explain how global developers share code similarly to pooling community labor.
- `[16:00 - 20:00]` **Conclusion & Call to Action:** Emphasize that a balanced community—whether on a farm or in a tech hub—requires the empowered, equal leadership of both men and women.

3. Modern Society Challenge: Food Security

TRI-LANGUAGE ABSTRACTS

English: Addressing the critical modern challenge of food security and supply chain inefficiency, this presentation proposes the implementation of digital agricultural tracking systems. By leveraging platforms like AgriTrack, we can optimize food distribution, reduce post-harvest losses, and ensure economic stability for rural farming communities.

Kiswahili: Kukabiliana na changamoto kuu ya kisasa ya usalama wa chakula na ukosefu wa ufanisi katika mfumo wa ugavi, wasilisho hili linapendekeza utekelezaji wa mifumo ya kidijitali ya ufuatiliaji wa kilimo. Kwa kutumia mifumo kama AgriTrack, tunaweza kuboresha usambazaji wa chakula, kupunguza hasara baada ya mavuno, na kuhakikisha utulivu wa kiuchumi kwa jamii za wakulima vijijini.

Kikuyu: Kumenyana na thina munene wa ugitiri wa irio na wagi wa mutaratara mwega wa kugayana indo, uhoro uyu urarehe mutaratara wa kuhuthira tekinoloji kumenya uria kilimo kiguthii. Kuhuthira mitaratara ta AgriTrack, no tuhote kugayana irio wega, kunyihia hasara thutha wa magetha, na gutigirira uikari wa kiuchumi kwi arimi a gicagi.

20-MINUTE VIDEO SCRIPT OUTLINE

- `[0:00 - 3:00]` **Introduction:** Define food security as a pressing modern challenge. Hook the viewer with statistics on food waste and rural poverty in Africa.
- `[3:00 - 10:00]` **Deep Dive/Core Analysis:** Break down the systemic inefficiencies causing the challenge: poor logistics, lack of real-time market data, and reliance on analog record-keeping.
- `[10:00 - 16:00]` **The Solution:** Present digital agricultural tracking as the cure. Detail how data-driven logistics and direct-to-buyer mobile apps optimize food distribution.
- `[16:00 - 20:00]` **Conclusion & Call to Action:** Reiterate that technology is the bridge to food security. Call on policymakers and tech developers to invest in rural digital infrastructure.